Pardeep Kumar

SCIENTIFIC SOFTWARE ENGINEER | C++ | PYTHON | JULIA | HIGH-PERFORMANCE COMPUTING

Amsterdam, the Netherlands | +31 6 1311 9813 | pardeep.iitb@gmail.com | Portfolio
LinkedIn | GitHub | Google Scholar | Languages: English, Hindi, Dutch (A2)

PROFILE

Scientific software engineer with more than ten years of experience developing numerical solvers, high-performance simulation software, optimisation algorithms, and engineering applications. Experience spans energy, semiconductors, CAE, finance, computational electromagnetics, CFD, and scientific research. My submitted doctoral thesis was developed at CWI Amsterdam and TU Delft.

Strong at translating mathematical and physical models into robust, tested, maintainable software, from formulation and implementation through profiling, interoperability, verification, and documentation.

SELECTED ENGINEERING IMPACT

30X	30 MIN TO 15 SEC	>15X
Faster constrained thermodynamic optimisation formulation	GPU-enabled scientific workflow after C++ and CUDA migration	Higher throughput for multi-client simulation data workflows

TECHNICAL STRENGTHS

Programming C++, Python, Julia, C#, Java, F#, MATLAB, TypeScript, and Erlang.	Numerical methods Finite-volume and finite-difference methods, nonlinear optimisation, Riemann solvers, thermodynamics, CFD, and numerical linear algebra.
High-performance computing CUDA, MPI, multithreading, concurrent programming, shared memory, profiling, and performance optimisation.	Software engineering Modular architecture, OOP and functional programming, MVC, API interoperability, unit testing, TDD, Git, Perforce, and CI-oriented development.
Computational physics Multiphase flow, electromagnetics, wave propagation, real-fluid thermodynamics, aeroacoustics, and numerical relativity.	AI and data workflows Neural surrogate models, scientific-data pipelines, automated training and validation, and reproducible experiments.

RECENT EXPERIENCE

PhD Researcher in Scientific Computing — Aug 2022-Aug 14, 2026

Centrum Wiskunde & Informatica (CWI), Amsterdam

- Developed Julia and Python solvers for transient multiphase, multicomponent flow with an emphasis on robustness, convergence, validation, and efficiency.
- Reformulated constrained phase-equilibrium algorithms, achieving approximately 30x speedup over a traditional nested approach.
- Built neural thermodynamics workflows and scientific software for hyperbolic flow models and numerical relativity.

Software Engineer — Feb-Aug 2022

ASM International, Almere

- Enhanced and refactored a Java simulator for semiconductor thin-film deposition equipment.
- Improved modularity, maintainability, testability, and incremental feature delivery.

Software Engineer — Jun 2019-Dec 2021

Shell, Amsterdam | Individual contributor

- Migrated Python workflows to C++ and CUDA, reducing a representative runtime from about 30 minutes to 15 seconds.
- Designed concurrent data infrastructure and C#/C++ interoperability for process and pipeline simulation tools.
- Improved multi-client throughput by more than 15x and built modular visualisation components.

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SCIENTIFIC SOFTWARE ENGINEERING EXPERIENCE AND OUTPUT

EARLIER ENGINEERING EXPERIENCE

Software Engineer **Feb-Apr 2019**

Aakraya Research, Bangalore | Team of 2

- Developed Python regression-model generators and post-trade performance-analysis tools.
- Built profit-and-loss analysis tools for evaluating live-trading performance.
- Improved C++ order-book infrastructure for efficient matching and market-data processing.

Software Engineer II **May-Nov 2018**

KLA-Tencor, Chennai | Team of 2

- Built C# and F# cross-application infrastructure and a C#/F#/MATLAB wafer-thickness simulator.
- Developed shared C++ and C# libraries for common functionality across server applications.
- Implemented unit tests and Erlang fault tolerance with failover and recovery.

Software Engineer **Mar 2016-Apr 2018**

Altair, Bangalore | Team of 6

- Developed C++ shared-memory infrastructure and geometry and mesh tools for HyperMesh.
- Modernised C++/Python bindings from SWIG to Boost.Python and built Python testing frameworks.
- Contributed to an MVC refactoring and extended COM-based application APIs.

Research and Development Engineer **Jul 2014-Mar 2016**

Fluidyn, Bangalore | Team of 4

- Developed an MPI-parallel finite-volume electromagnetic solver in C++ and C#.
- Implemented second- and third-order spatial schemes and near-to-far-field radar cross-section processing.
- Extended a Navier-Stokes solver for far-field aeroacoustic prediction using retarded-time integration.

EDUCATION

PhD in Mechanical Engineering **2022-2026**

TU Delft | Research conducted at CWI Amsterdam

Thesis submitted; defence planned for January 2027. Numerical methods for multicomponent, multiphase pipeline transport.

MSc in Aerospace Engineering **2012-2014**

Indian Institute of Technology Bombay, India

Finite-volume time-domain methods for electromagnetic propagation and scattering.

SELECTED PEER-REVIEWED OUTPUT

1. P. Kumar and P. I. Rosen Esquivel. A Reformulation of UVN-Flash for Multicomponent Two-Phase Systems with Application to CO2-Rich Mixture Transport in Pipelines. Computers & Fluids, 314, 107108, 2026.
2. P. Kumar and P. I. Rosen Esquivel. Solving the UVN-Flash Problem in TVN-Space. Fluid Phase Equilibria, 599, 114528, 2026.
3. P. Kumar, B. Sanderse, P. I. Rosen Esquivel, and R. A. W. M. Henkes. A New Temperature Evolution Equation That Enforces Thermodynamic Vapour-Liquid Equilibrium in Multiphase Flows. Computers & Fluids, 289, 106524, 2025. View publications.

INDEPENDENT PRODUCT WORK

Built and deployed four public web products using TypeScript, React, Next.js, Python, Rust, PostgreSQL, and cloud platforms. View web applications.